

Maddox Gonzalez

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EDUCATION

University of Central Florida

Bachelor of Science in Mechanical Engineering

Pursuing Masters in Aerospace Thermofluids through BS-MS accelerated program

Distinctions: Dean's List | President's Honor Roll

Expected Graduation: Fall 2026

GPA: 3.565

WORK EXPERIENCE

UCF Undergraduate Learning Assistant

Jun 2025 – Present

Instructor

- Guiding students through coursework, clarifying key concepts, and promoting independent thinking for college algebra through Calculus I.

Undergraduate Teaching Assistant (Fluid Mechanics)

Jan. 2026 – April. 2026

Teacher's Assistant

- Support the professor through grading, creating Ansys Fluent projects, and assisting students with fluid mechanics concepts and problem solving.

TECHNICAL SKILLS & CERTIFICATIONS

SolidWorks Associate | Certified Fusion 360 User | FE Mech. Exam (passed) | C | Python | MatLab | ANSYS Fluent | ANSYS Mech

PROJECTS & RESEARCH

Functionally Graded Resin Infused Ceramic Thermal Protection System

May. 2026 - Present

Undergraduate Research - Composite Materials & Structures Lab

- Generated and developed a functionally graded ceramic thermal protection system (TPS) for hypersonic vehicles featuring interconnected vascular channels to enable phase-change cooling for hypersonic applications.
- Planned resin infusion to evaluate structural integrity and thermal response, followed by torch testing to benchmark performance against commercial thermal protection systems (TPS) under high-heat flux conditions.

Hybrid Aerial Underwater Drone

May. 2026 - Present

Undergraduate Capstone Project

- Designing a hybrid aerial-underwater drone for coral reef research, currently performing system-level function and component decomposition while defining operational and performance requirements for multi-environment mobility and data collection.

Thermal Modeling of Phase Change Using Neural Networks

Feb. 2026 - April. 2026

Research Project

- Implemented a physics-informed neural network (PINN) to model 1-D phase-change solidification by embedding energy conservation and boundary conditions into the loss function.
- Achieved relative errors below 1×10^{-3} using hard boundary-condition enforcement, reproducing published benchmark results and outperforming finite element solutions.

Wind Tunnel

Jan. 2026 - Present

Personal Project

- Modeling a small-scale wind tunnel to study airflow characteristics and measure velocity, drag, and pressure on test airfoils.
- Utilizing ANSYS Fluent to simulate flow conditions and compare experimental results against theoretical results.

PROFESSIONAL DEVELOPMENT

SolidWorks Student Leadership Program

Sep. 2025 - Present

Student Organization

- Participating in workshops and collaborative assignments to enhance proficiency in SolidWorks. I am a Certified SolidWorks Associate, and am working towards becoming a Certified SolidWorks Professional.

NASA L'SPACE Mission Concept Academy

Aug. 2024 - Dec. 2024

Team Project - Lead Engineer

- Collaborated on a hypothetical team mission task to identify and evaluate lunar pits for potential thermal isolation and structural stability as candidates for long-term lunar habitation.
- Designed Thermal Management & Power Systems for the spacecraft to ensure mission requirements are met.

Knights Experimental Rocketry (KXR)

Aug. 2023 – May. 2024

Member of HPR Recovery Team & IREC Aerostructures Team

- Oversaw the recovery phase of the club rocket with a small team, including apogee separation and parachute deployments.
- Analyzed and optimized rocket components, such as: fins, materials, and nose-cone shape to enhance aerodynamic performance and stability.